Brain vs LLM

Where do they actually resemble each other?

01

How the Brain Works

02

How LLMs Work

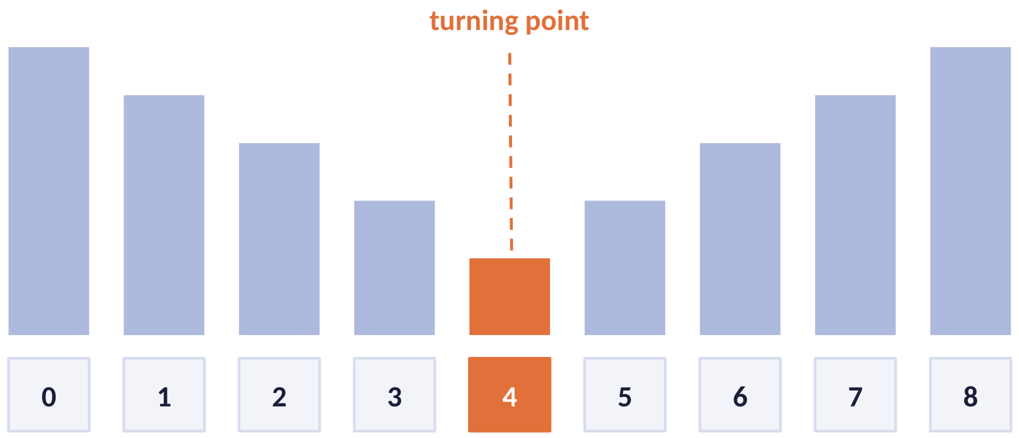
03

The Resemblance

Starting with the brain — how it recognizes patterns and solves problems

Recognizing a Pattern

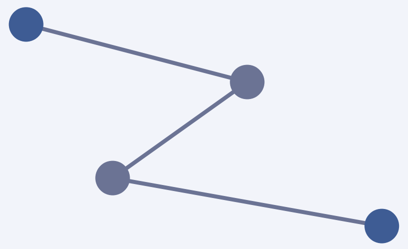
Example: find where a sequence stops decreasing and starts increasing



- With a teacher, you walk through it step by step
- With practice, you recognize it instantly
- Solving similar problems trains your brain to spot the shape faster

BEHIND THE SCENES

First time — long chain



After practice — one strong link

Neurons that fire together wire together — repetition strengthens the connection.

Building a Pattern Library

More practice grows your library — and shapes the brain itself

Greedy → Binary Search → Greedy + Binary Search → Hybrid (+ Trees)

Basic problems train single techniques — harder ones combine what you already know

Is the Brain an ML Model?

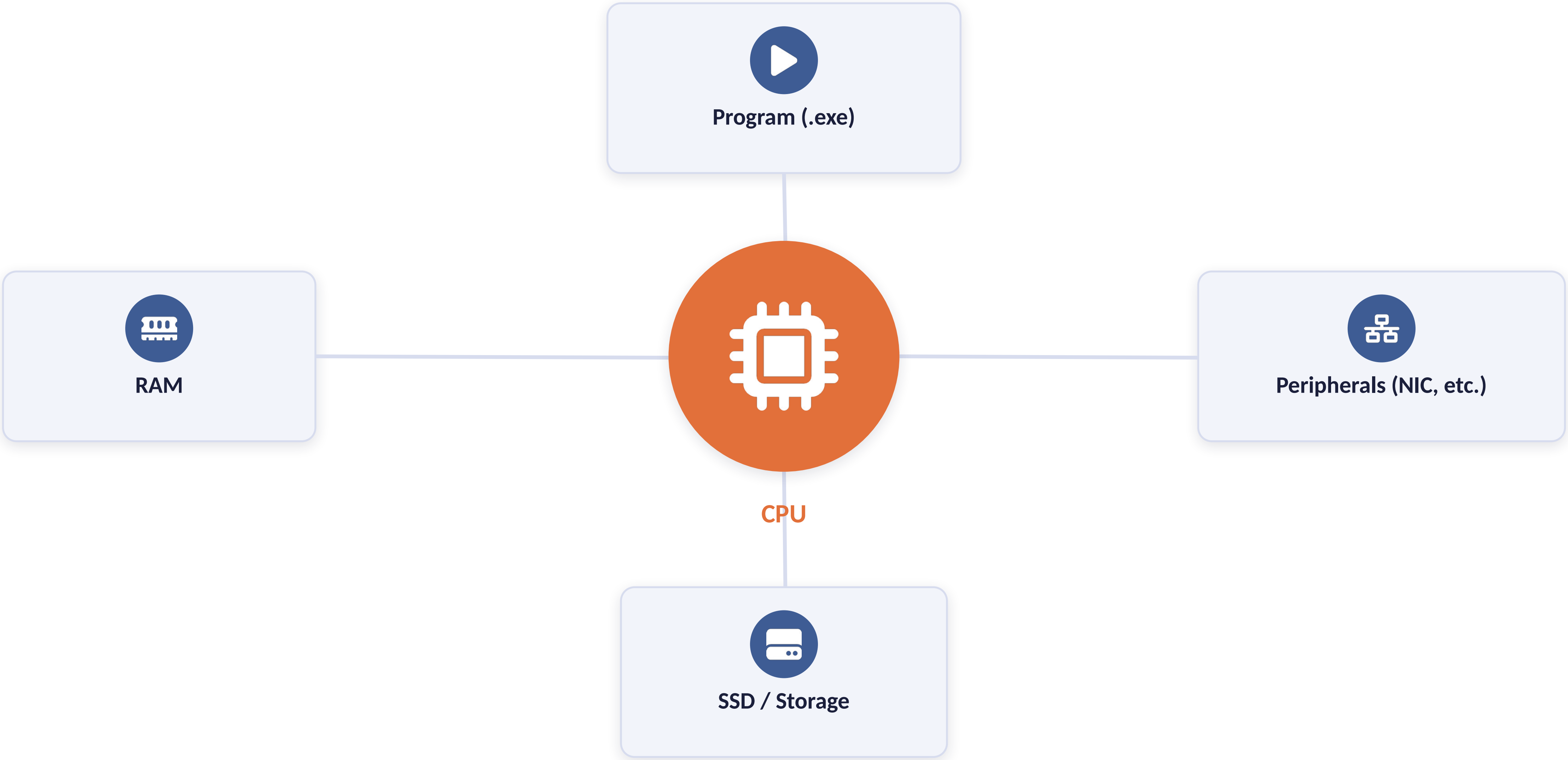
Not exactly — but close in one way.

During sleep, weak or unused connections between neurons fade, while the ones you use again and again grow stronger.

That strengthening is the closest thing to a “weight” in the brain.

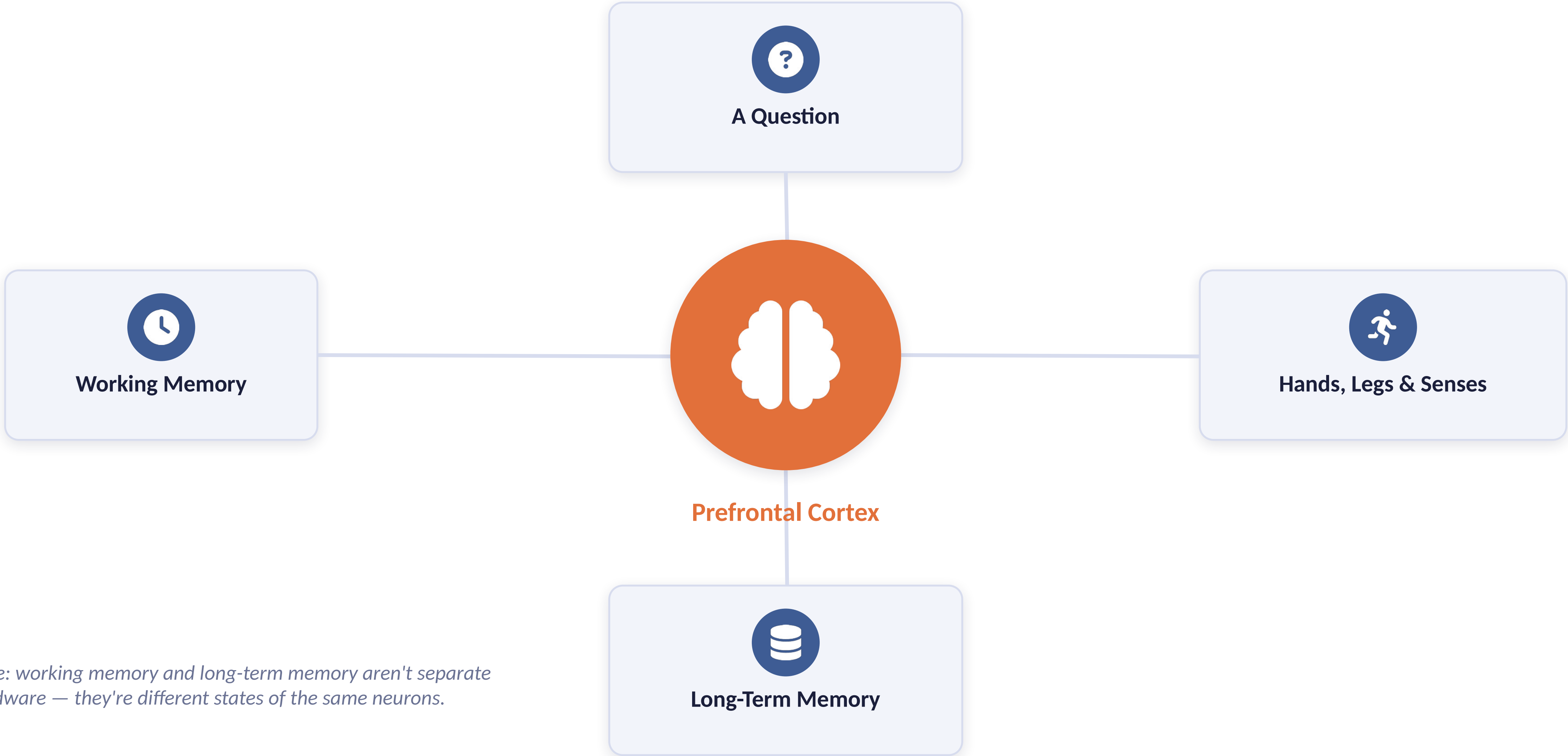
How a Computer Runs a Program

When you execute a program, the CPU pulls in everything else as needed



The Brain's Version

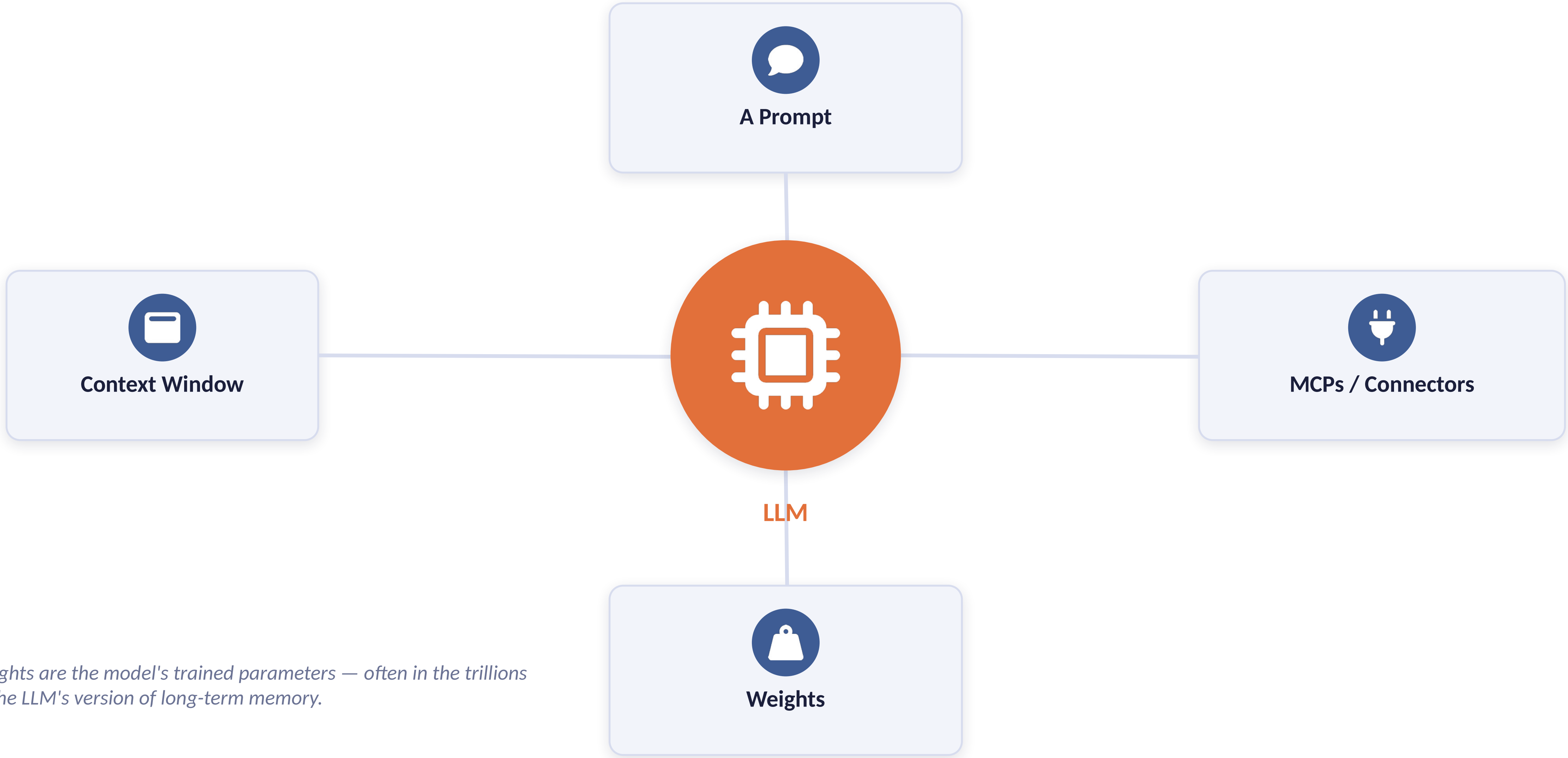
The prefrontal cortex pulls in memory and the senses to answer



Note: working memory and long-term memory aren't separate hardware — they're different states of the same neurons.

The LLM Version

The model pulls in context, tools, and its trained weights to respond



Weights are the model's trained parameters — often in the trillions — the LLM's version of long-term memory.

Brain vs Machine: The Efficiency Gap

Even without trillions of parameters, the brain still holds real advantages

10×

more power a model takes compared to the brain

Room Temp

the brain runs cool — GPUs need active cooling to keep from overheating

~5×

more computations the brain performs, using far less energy to do it

Not All Fire

only the neurons needed for a task activate — most stay quiet

Even running on far less power, the brain still does more — and does it more efficiently.

How an ML Model Learns

Four pieces that turn data into a working model

GOAL

Train the model to predict correctly

UTILITY

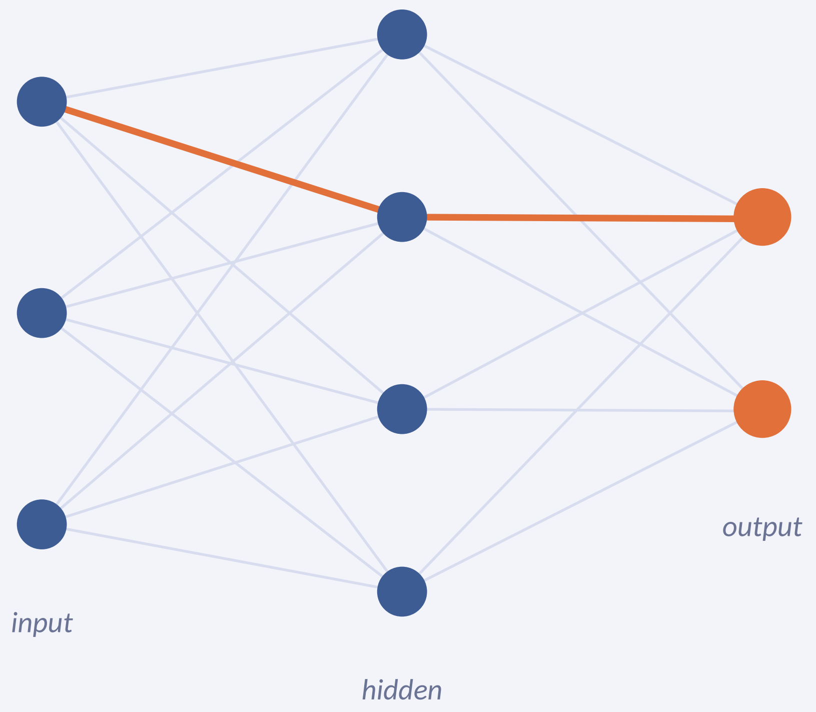
Minimize the error — how wrong the predictions are

REGULARIZATION

Keeps weights working on new data, not just memorized training data

GRADIENT

The signal that nudges the weights, step by step, toward less error



gradient flows back to update weights

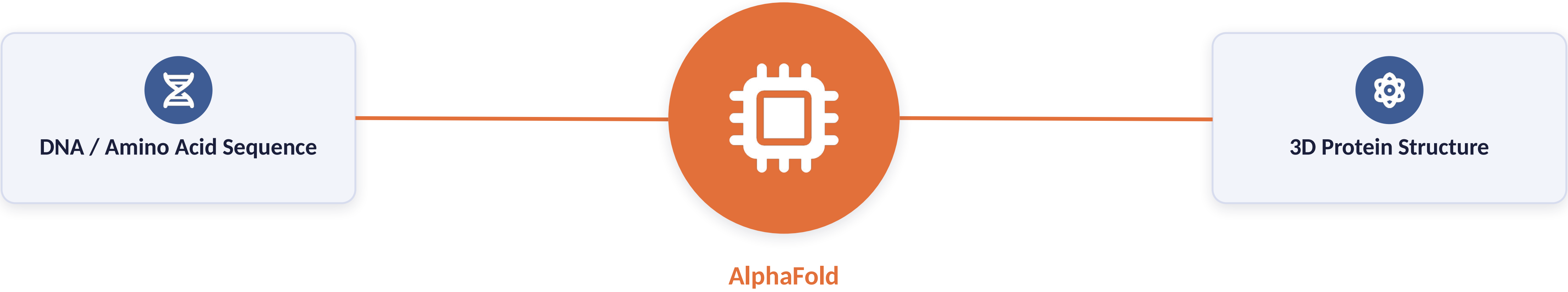
Pulling an All-Nighter: The Brain as a Model

Same four pieces, mapped onto exam night

GOAL	**Score well on the exam**	Predict the right answers under test conditions
UTILITY / ERROR	**Marks lost**	Points you could have gotten right, but didn't
REGULARIZATION	**Caffeine ÷ (sleep deprivation + adrenaline)**	What keeps you performing on the actual exam, not just on what you memorized
GRADIENT	**Knowledge absorbed per hour left**	How much you can actually take in, given the time remaining before the exam

AlphaFold

An AI system by DeepMind that predicts a protein's 3D shape from its sequence



Developed by DeepMind (Google), AlphaFold solved a problem biologists had worked on for over 50 years — predicting how a protein folds just from its sequence.

Talks & Further Reading

For anyone who wants to go deeper

TALKS

- How LLMs Learn Their Weights — Training at Scale
  Stanford CS229N: Building Large Language Models
- Another AI Talk Worth Watching

DOCUMENTS I WROTE

- Overview of AI
- Consciousness of AI
- LLMs vs Humanity

MORE TO EXPLORE

INDUSTRY / APPLIED

Skills · Agents · Loops · Graphical Loops · Harnessing · MCPs · Prompt Injection · Jailbreaking · Graph Engineering

RESEARCH

Interpretability · Distillation · RSI (Recursive Self-Improvement) · ...and more

Ongoing areas of work across the industry and research — this list keeps growing